SRM INSTITUTE OF SCIENCE & TECHNOLOGY, NCR CAMPUS

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# [Stock Market Portfolio]

## A PROJECT REPORT

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of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

# SRM INSTITUTE OF SCIENCE & TECHNOLOGY

(Under Section 3 of UGC Act, 1956)

## BONAFIDE CERTIFICATE

Certified that this report titled "**Stock Market Portfolio**" is the Bonafide work of " **Utkarsh Kumar Srivastava[RA2411030030059], Vibhanshu Mittal[RA2411030030062] and Aditya Agrahari[RA2411030030044]**", who carried out this work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other report on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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**Index**


| Chapters. | Topic | Page No. |
|---|---|---|
| 1 | Introduction | 4 |
| 2 | Project Description | 5-6 |
| 3 | Programming | 7-20 |
| 4 | Results | 21-25 |
| 5 | Conclusion | 26 |
| 6 | References | 27 |

# Chapter 1 : INTRODUCTION

¬ **Main Idea:**
My project simulates a **Stock Market Management System** using DBMS concepts. It provides features like:
¬ User authentication (via email and password)
¬ Admin and normal user roles
¬ Stock listing, buying, selling, and portfolio management
¬ Transaction history tracking
¬ Wallet/balance management system
¬ Real-time stock price updates (simulated)
¬ Database-driven secure data storage and retrieval
The stock market is one of the most important financial systems in the modern economy, where investors buy and sell shares of companies to grow their wealth. This system helps users to manage their investments efficiently by allowing them to view available stocks, purchase shares, sell holdings, and monitor their portfolio performance without relying on complex real-world trading platforms. It simplifies stock market operations for learning and practical understanding.
The project allows users to securely log in, maintain a virtual wallet, invest in stocks, and track their transaction history. The admin can manage stock listings, update prices, and monitor overall system activity.
The main objective of this project is to design a structured, secure, and efficient **Stock Management System using DBMS**. It focuses on proper database design, including tables, relationships, primary and foreign keys, and data integrity constraints.
By developing this system, we gain practical knowledge of database operations such as CRUD (Create, Read, Update, Delete), normalization, and query optimization using SQL.
Simulating a stock market system is a valuable academic exercise that strengthens understanding of DBMS concepts and real-world financial systems. In summary, this project demonstrates how database systems can be effectively used to manage complex financial data and transactions efficiently.

# Chapter 2 :PROJECT DESCRIPTION

The **Stock Market Management System** is designed to perform essential stock trading operations in a simulated and simplified manner. The application operates through a database-driven interface and uses a relational DBMS for efficient data storage and management.

¬ **System Overview**

The program consists of two roles – **User** and **Admin**. A user can perform operations such as viewing stocks, buying and selling shares, managing their portfolio, and checking transaction history.

An admin, on the other hand, can manage stock listings, update stock prices, and monitor overall system activities for proper control and security.

**Modules:**

1. **Login Module** – Validates user credentials and ensures secure access.
2. **User Module** – Handles stock buying, selling, portfolio management, wallet balance, and transaction history.
3. **Admin Module** – Manages stock data, updates prices, and controls user activities.
4. **Database Management Module** – Uses DBMS tables (users, stocks, transactions, portfolio, wallet) for persistent data storage.

**Functional Requirements:**

- Secure authentication using email and password.
- Maintain user wallet balance and portfolio records.
- Enable stock buying and selling functionality.
- Record all transactions with proper details.
- Admin control over stock listings and updates.
- Ensure relational database integrity using primary and foreign keys.

**Non-Functional Requirements:**

- User-friendly interface (GUI-based).
- High data accuracy and consistency.
- Efficient query processing and transaction handling.
- Error handling for invalid operations (e.g., insufficient balance).
- Secure and reliable data storage.

**Hardware and Software Requirements:**

- **Software:** MySQL / PostgreSQL, Java , DBMS tools (e.g., MySQL Workbench)
- **Hardware:** Minimum 4GB RAM, 500MB disk space

**Key technologies and design aspects include:**

• **Database System:** Relational DBMS (MySQL/PostgreSQL) for structured data storage.

• **Query Language:** SQL for data manipulation and retrieval.

• **Data Structures:** Tables with relationships (Primary Keys, Foreign Keys).

• **Normalization:** Ensures minimal redundancy and efficient data organization.

• **Object-Oriented Design (implemented with Java):** Classes represent entities like User, Stock, Transaction, and Portfolio.
• **Transaction Management:** Ensures consistency and reliability during buy/sell operations.
This project demonstrates how database systems can be used to simulate real-world stock market operations efficiently while maintaining data integrity, security, and performance.

# Chapter 3 :PROGRAMMING

```
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.Statement;
import java.util.Scanner;

public class AddStock {
   public static void add() {
      try(Scanner sc = new Scanner(System.in)){
      System.out.println("Enter the stock symbol:");
      String symbol = sc.nextLine();
      System.out.println("Enter Stock name:");
      String sname = sc.nextLine();
      System.out.println("Enter Initial Stock Price:");
      double price = 0;
      try {
         price = Double.parseDouble(sc.nextLine());
      } catch (NumberFormatException e) {
         System.out.println("Invalid price. Defaulting to ₹0.0");
      }

      String sql = "INSERT INTO stocks (symbol, name, price) VALUES (?, ?, ?)";
      try (Connection con = DBConnection.getConnection()) {
         String createTableSQL = "CREATE TABLE IF NOT EXISTS stocks (id INT
AUTO_INCREMENT PRIMARY KEY, symbol VARCHAR(50) UNIQUE, name
VARCHAR(255), price DOUBLE)";
         try (Statement stmt = con.createStatement()) {
            stmt.execute(createTableSQL);
         }
         try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setString(1, symbol.toUpperCase());
            ps.setString(2, sname);
            ps.setDouble(3, price);
            ps.executeUpdate();
            System.out.println(
                  "Stock '" + sname + "' (" + symbol.toUpperCase() + ") added to
database successfully!");
         }
      } catch (Exception e) {
         System.out.println("Error adding stock. Does the symbol already exist?");
      }
```

```
        }
    }
}


    // ---------------- Admin Login ----------------
    import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.Statement;
import java.util.Scanner;

class adminLogin
{
    public static void LoginAdmin()
    {
        try(Scanner sc = new Scanner(System.in)){
        String name = "ADMIN";
        String email = "admin@gmail.com";
        String password = "admin";
        String pass = password;
        String enc ="";
        for(int i=0;i<pass.length();i++)
        {
            char ch = pass.charAt(i);
            ch += 15;
            enc += ch;
        }
        System.out.println("Enter Admin email");
        String e = sc.nextLine();
        System.out.println("Enter Admin password");
        String p = sc.nextLine();
        System.out.println();

        String createTableSQL = "CREATE TABLE IF NOT EXISTS admins (id INT
AUTO_INCREMENT PRIMARY KEY, name VARCHAR(255), email
VARCHAR(255) UNIQUE, password VARCHAR(255))";
        String sql = "INSERT IGNORE INTO admins (name, email, password)
VALUES (?, ?, ?)";

        try (Connection con = DBConnection.getConnection())
        {
            try (Statement stmt = con.createStatement()) {
                stmt.execute(createTableSQL);
                try {
                    stmt.execute("ALTER TABLE admins ADD COLUMN name
```

```
VARCHAR(255)");
            } catch(Exception ignored) {}
        }
        try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setString(1, name);
            ps.setString(2, email);
            ps.setString(3, enc);
            ps.executeUpdate();
        }
    }
    catch(Exception e1)
    {
        e1.printStackTrace();
    }

    boolean flag = true;
    while(flag == true)
    {
        if((e.equals(email))&&(p.equals(password)))
        {
            System.out.println("Admin was Login Successfully");
            break;
        }
        else
        {
            System.out.println("Wrong credentials: Re-enter email and password");
        }
    }
    }
  }
}

  // ---------------- Database Connection ----------------
  import java.sql.DriverManager;
import java.sql.Connection;

class DBConnection
{
  static String url = "jdbc:mysql://localhost:3306/stocks";
  static String name = "root";
  static String password = "Uk@17122006";


  public static Connection getConnection()
  {
```

```
        try
        {
            Connection con = DriverManager.getConnection(url,name,password);
            return con;
        }
        catch(Exception e)
        {
            System.out.println("Error Occured");
            e.printStackTrace();
            return null;
        }
    }
}

    // ---------------- Main Menu ----------------
public static void mainMenu()
    {
        System.out.println();
        System.out.println();
        System.out.println("""
        ==========================================
          STOCK MARKET PORTFOLIO MANAGEMENT
        ==========================================

        1. Register New User
        2. User Login
        3. Admin Login
        4. Exit
        """);
    }
    public static void userMenu()
    {
        System.out.println();
        System.out.println();
        System.out.println("""
        ==========================================
                  USER DASHBOARD
        ==========================================

        1. Add Funds to Wallet
        2. View Wallet Balance
        3. Buy Stocks
        4. Sell Stocks
        5. View Profit / Loss
        6. View Transaction History
```

```
        7. View Live Stock Price
        8. View Stock Details
        9. View your stocks
        10. Logout
        """);
    }
    public static void adminMenu()
    {
        System.out.println();
        System.out.println();
        System.out.println("""
        ==========================================
                    ADMIN PANEL
        ==========================================

        1. Add New Stock
        2. Update Stock Details
        3. Remove Stock
        4. View All Stocks
        5. View All Users
        6. View All Database Data
        7. Logout
        """);
    }
}

    // ---------------- User Login ----------------
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

class userLogin
{
    public static String currentUserEmail = null;
    public static void LoginUser()
    {
        try(Scanner sc = new Scanner(System.in)){

        while (true)
        {
            System.out.println("Enter email");
            String email = sc.nextLine();
            System.out.println("Enter password: ");
            String password = sc.nextLine();
```

```
            String sql = "SELECT * FROM users WHERE email=? AND password = ?";

            String decrypt = "";
            int l = password.length();
            for(int i=0;i<l;i++)
            {
               char ch = password.charAt(i);
               ch += 10;
               decrypt += ch;
            }
            try(Connection con = DBConnection.getConnection();
               PreparedStatement ps = con.prepareStatement(sql))
            {
               ps.setString(1, email);
               ps.setString(2, decrypt);

               ResultSet rs = ps.executeQuery();

               if(rs.next())
               {
                  System.out.println("Login Successful");
                  currentUserEmail = email;
                  break;
               }
               else
               {
                  System.out.println("ERROR: Email or password was wrong");
               }
            }
            catch(Exception e)
            {
               System.out.println("Login Error");
               e.printStackTrace();
            }
         }
         }

      }
   }

   // ---------------- View Wallet Balance ----------------
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
```

```
import java.util.Scanner;

public class ViewWalletBalance {
   public static void view() {
      try(Scanner sc = new Scanner(System.in)){
      System.out.println("Enter your wallet password:");
      String pass = sc.nextLine();

      System.out.println("\n--- Your Wallet Balance ---");
      String sql = "SELECT balance FROM wallet WHERE password = ?";

      try (Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql)) {

         ps.setString(1, pass);
         ResultSet rs = ps.executeQuery();

         if (rs.next()) {
            System.out.println("Balance: ₹" + rs.getDouble("balance"));
         } else {
            System.out.println("No wallet found with that password.");
         }
      } catch (Exception e) {
         System.out.println("Error fetching wallet balance.");
      }
      System.out.println("--------------------------\n");
      }
   }
}


   // ---------------- View Transaction History ----------------
   import java.util.Scanner;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

public class ViewTransactionHistory {
   public static void view() {
      try(Scanner sc = new Scanner(System.in)){
      System.out.println("Enter your wallet password to view history:");
      String pass = sc.nextLine();

      System.out.println("\n--- Transaction History ---");
      String sql = "SELECT * FROM transactions WHERE wallet_password = ?
```

```
ORDER BY date DESC";

        try (Connection con = DBConnection.getConnection();
                PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, pass);
            ResultSet rs = ps.executeQuery();

            boolean found = false;
            while (rs.next()) {
                System.out.println(
                        rs.getTimestamp("date") + " | " + rs.getString("type") + " | ₹" +
rs.getDouble("amount"));
                found = true;
            }

            if (!found) {
                System.out.println("No recent transactions.");
            }
        } catch (Exception e) {
            System.out.println("Database error or table missing (have you made a deposit
yet?).");
        }
        System.out.println("--------------------------\n");
        }
    }
}



    // ---------------- Database Table ----------------

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;

public class DatabaseDumper {
    public static void dumpDatabase() {
        System.out.println("ENTIRE DATABASE DUMP ");

        // Known tables in your schema. Add more if you create new ones!
        String[] tables = {"users", "admins", "wallet", "transactions", "stocks",
"user_portfolio"};
```

```
        try (Connection con = DBConnection.getConnection()) {
            if (con == null) {
                System.out.println("Database connection failed.");
                return;
            }

            for (String table : tables) {
                System.out.println("\n--- Table: " + table.toUpperCase() + " ---");
                try {
                    String sql = "SELECT * FROM " + table;
                    try (PreparedStatement ps = con.prepareStatement(sql);
                         ResultSet rs = ps.executeQuery()) {

                        ResultSetMetaData rsmd = rs.getMetaData();
                        int columnCount = rsmd.getColumnCount();

                        // Print headers
                        for (int i = 1; i <= columnCount; i++) {
                            System.out.print(String.format("%-25s",
rsmd.getColumnName(i)));
                        }
                        System.out.println();

                        // Print divider
                        for (int i = 1; i <= columnCount; i++) {
                            System.out.print("-------------------------");
                        }
                        System.out.println();

                        // Print rows
                        boolean hasData = false;
                        while (rs.next()) {
                            hasData = true;
                            for (int i = 1; i <= columnCount; i++) {
                                String val = rs.getString(i);
                                if (val == null) val = "NULL";
                                if (val.length() > 24) val = val.substring(0, 21) + "...";
                                System.out.print(String.format("%-25s", val));
                            }
                            System.out.println();
                        }

                        if (!hasData) {
                            System.out.println("(Table is empty)");
                        }
```

```
                }
            } catch (Exception e) {
                System.out.println("Table '" + table + "' does not exist (yet).");
            }
        }
        System.out.println("\n========= END OF DATABASE DUMP
=========");

    } catch (Exception e) {
        System.out.println("Error connecting to database.");
    }
  }
}



  // ---------------- View All Stocks ----------------

  import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Statement;

public class ViewAllStocks {
  public static void view() {
    System.out.println("\n--- All Stocks in System ---");

    String createTableSQL = "CREATE TABLE IF NOT EXISTS stocks (id INT
AUTO_INCREMENT PRIMARY KEY, symbol VARCHAR(50) UNIQUE, name
VARCHAR(255), price DOUBLE)";
    String sql = "SELECT * FROM stocks";

    try (Connection con = DBConnection.getConnection()) {
        // Ensure table exists
        try (Statement stmt = con.createStatement()) {
            stmt.execute(createTableSQL);
        }

        try (PreparedStatement ps = con.prepareStatement(sql);
                ResultSet rs = ps.executeQuery()) {

            boolean found = false;
            while (rs.next()) {
                System.out.println("Symbol: " + rs.getString("symbol") +
                        " | Name: " + rs.getString("name") +
```

```
                " | Price: ₹" + rs.getDouble("price"));
            found = true;
        }

        if (!found) {
            System.out.println("No stocks currently exist in the system.");
        }
    }
} catch (Exception e) {
    System.out.println("Error fetching stocks.");
    e.printStackTrace();
}
System.out.println("----------------------------\n");
}
}



// ---------------- Creating Wallet ----------------
class Wallet {
static Scanner sc = new Scanner(System.in);

public static void createWallet() {
    System.out.println("Enter your password: ");
    String pass = sc.nextLine();

    String createTableSQL = "CREATE TABLE IF NOT EXISTS wallet (id INT
AUTO_INCREMENT PRIMARY KEY, password VARCHAR(255), balance
DOUBLE)";
    String sql = "INSERT INTO wallet (password, balance) VALUES (?,?)";

    try (Connection con = DBConnection.getConnection();
            Statement stmt = con.createStatement()) {
        // Auto-create the table if it's missing
        stmt.execute(createTableSQL);

        try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setString(1, pass);
            ps.setDouble(2, 0.0);

            ps.executeUpdate();
            System.out.println("Wallet created successfully!");
        }
    } catch (Exception e) {
        System.out.println("Error Occurred");
```

```
            e.printStackTrace();
        }
    }


    // ---------------- Register New User ----------------
    public static void Register()
    {
        try(Scanner sc = new Scanner(System.in)){
        System.out.println("Enter Name:");
        String name = sc.nextLine();
        System.out.println("Enter email");
        String email = sc.nextLine();
        boolean flag = true;

        while(flag == true)
        {
            System.out.println("Enter a Strong password");
            String pass = sc.nextLine();
            System.out.println();
            System.out.println("Confirm Password");
            String pass1 = sc.nextLine();

            if(pass.equals(pass1))
            {
                String encrypt = "";
                int l = pass.length();
                for(int i=0;i<l;i++)
                {
                    char ch = pass.charAt(i);
                    ch += 10;
                    encrypt +=ch;
                }

                String createTableSQL = "CREATE TABLE IF NOT EXISTS users (id
INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(255), email
VARCHAR(255) UNIQUE, password VARCHAR(255))";
                String sql = "INSERT INTO users (name, email, password) VALUES
(?,?,?)";
                try(Connection con = DBConnection.getConnection())
                {
                    if (con == null)
                    {
                        System.out.println("DB connection failed");
                        return;
```

```
            }

            try (Statement stmt = con.createStatement()) {
                stmt.execute(createTableSQL);
                try {
                    stmt.execute("ALTER TABLE users ADD COLUMN name
VARCHAR(255)");
                } catch(Exception ignored) {}
            }

            try (PreparedStatement ps = con.prepareStatement(sql)) {
                ps.setString(1, name);
                ps.setString(2, email);
                ps.setString(3, encrypt);
                ps.executeUpdate();
                System.out.println("User Registered Successfully ");
                flag = false;
            }
        }
        catch(Exception e)
        {
            System.out.println("User was not registered!! Sorry");
            e.printStackTrace();
        }
    }
    else
    {
        System.out.println("Password doesn't match: Re-enter the password dude");
    }


  // ---------------- View All Users ----------------
  public static void view() {
      System.out.println("\n--- All Registered Users ---");
      try (Connection con = DBConnection.getConnection();
            java.sql.Statement stmt = con.createStatement()) {

        String sql = "SELECT u.id, u.name, u.email, COALESCE(w.balance, 0.0) as
balance FROM users u LEFT JOIN wallet w ON u.email = w.email";
        ResultSet rs = stmt.executeQuery(sql);

        boolean found = false;
        System.out.printf("%-5s | %-15s | %-22s | %-10s | %-10s | %-10s\n", "ID",
"Name", "Email", "Balance",
            "Portfolio", "Net Worth");
```

```
            System.out.println(
                    "-----------------------------------------------------------------------------------------
------------");
            while (rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
                String email = rs.getString("email");
                if (name == null || name.trim().isEmpty())
                    name = email;
                double balance = rs.getDouble("balance");
                double portVal = 0.0;

                String pSql = "SELECT SUM(p.quantity * s.price) as total FROM
user_portfolio p JOIN stocks s ON p.symbol = s.symbol WHERE p.email = ?";
                try (PreparedStatement psPort = con.prepareStatement(pSql)) {
                    psPort.setString(1, email);
                    ResultSet rsP = psPort.executeQuery();
                    if (rsP.next())
                        portVal = rsP.getDouble("total");
                } catch (Exception ignored) {
                }

                System.out.printf("%-5d | %-15s | %-22s | ₹%-9.2f | ₹%-9.2f | ₹%-9.2f\n",
                        id, name, email, balance, portVal, (balance + portVal));
                found = true;
            }
            if (!found) {
                System.out.println("No users found in database.");
            }
        } catch (Exception e) {
            System.out.println("Error fetching users or table not found.");
            // e.printStackTrace();
        }
        System.out.println("---------------------------\n");
    }
}
```

# Chapter 4 :RESULTS

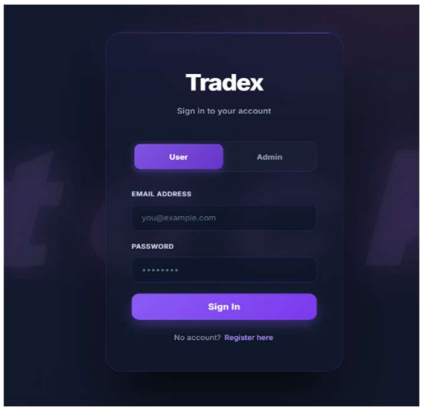



**Fig. 1 Register Users**

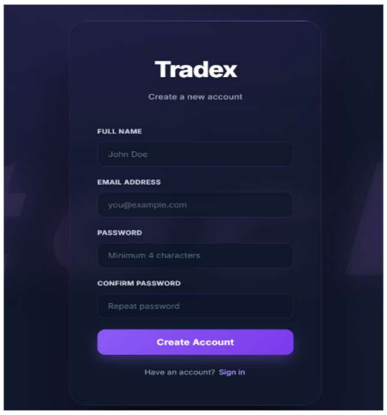
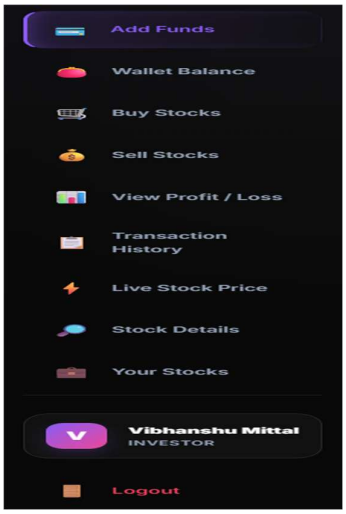
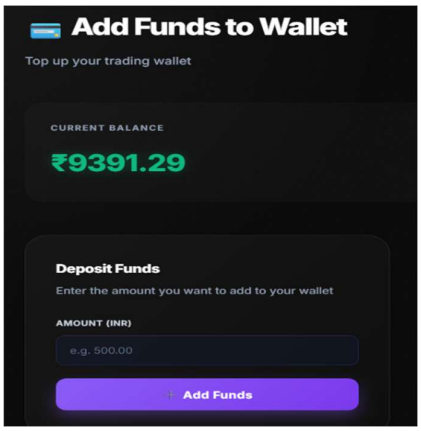


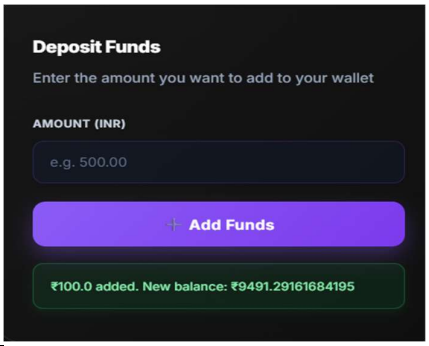

**Fig. 2 User menu Option**



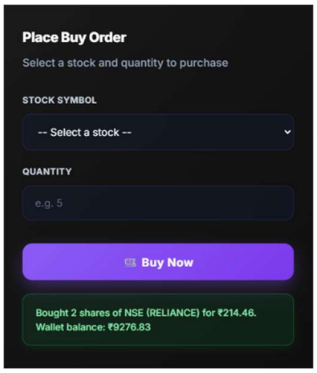
## Fig. 3 Balance Inquiry




## Fig. 4 Money Deposited




## Fig. 5 Buying Stocks




## Fig. 6 Transaction History

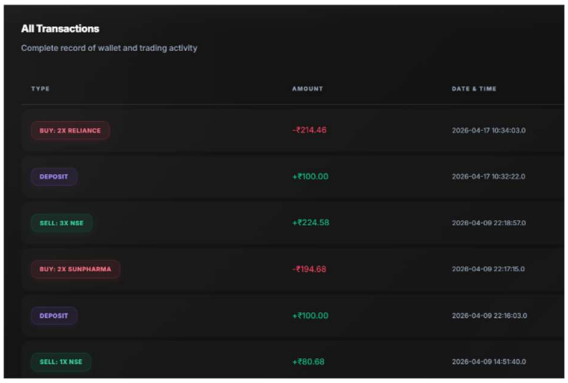



**Fig. 7 Viewing all Stocks**

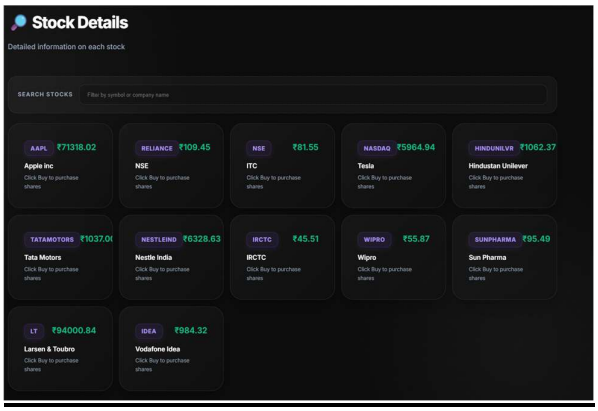
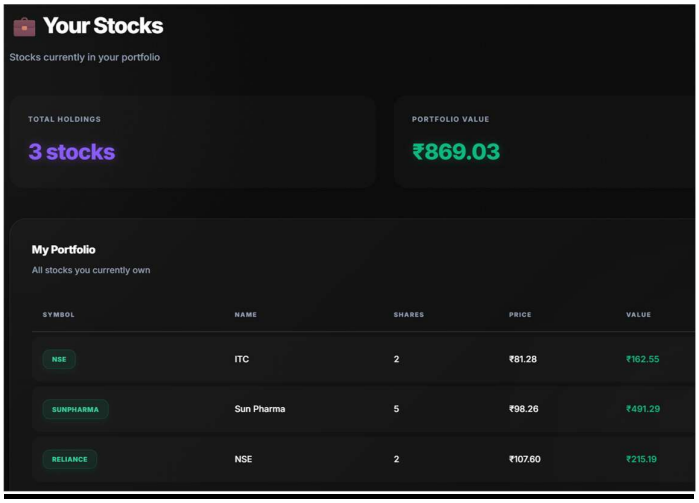
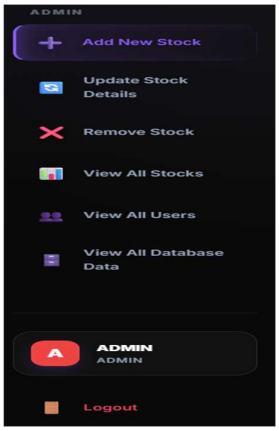


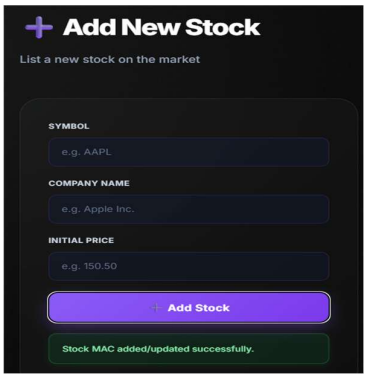

**Fig. 8 View your Stocks**



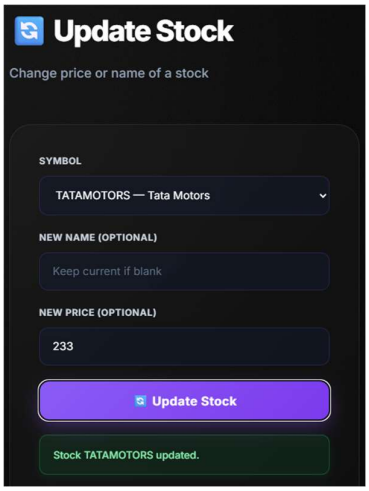
## Fig. 9 Admin Menu



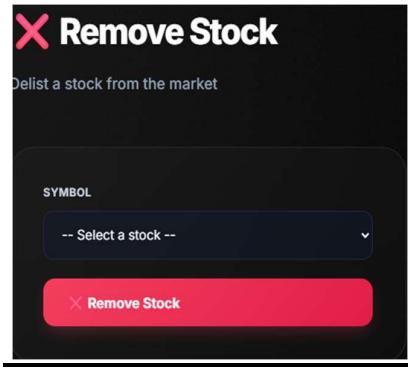

## Fig. 10 Add New Stocks



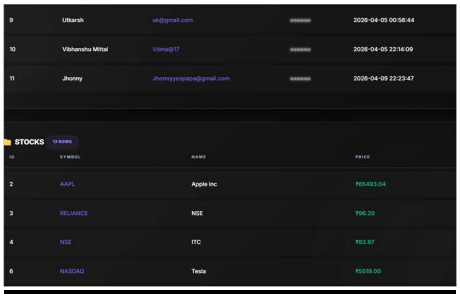

## Fig. 11 Update Stock Details



## Fig. 12 Removing Stock Detail from DB




## Fig. 13 All DB details in Admin



# Chapter 5 : CONCLUSION

The **Stock Market Management System Project** has been successfully developed and implemented using DBMS concepts.

It fulfills all the essential functions of a basic stock trading system, including user authentication, stock viewing, buying and selling of shares, portfolio management, wallet balance handling, and transaction history tracking.

The project demonstrates how database management concepts such as tables, relationships, primary and foreign keys, and normalization can be applied to simulate real-world financial systems. Through this project, practical knowledge of SQL queries, data organization, and database design has been gained.

The system operates efficiently, handling both user-level and admin-level operations accurately, while maintaining proper transaction records and ensuring data consistency. It also ensures data security by validating user credentials and maintaining integrity constraints within the database.

Overall, this project provided valuable experience in building a complete database-driven application that combines structured data management with logical operations. It can be further enhanced in the future by integrating real-time stock APIs, advanced data visualization dashboards, and a graphical user interface to make it more interactive and closer to real-world stock trading platforms.

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